

Exhibit 40

McKinsey & Company Report - Technology Trends Outlook 2022

McKinsey Technology Trends Outlook 2022

Report

August 2022



Introduction

by Michael Chui, Roger Roberts, and Lareina Yee

Technology continues to be a primary catalyst for change in the world. Technology advances give businesses, governments, and social-sector institutions more possibilities to lift their productivity, invent or reinvent offerings, and contribute to humanity's well-being. And while it remains difficult to predict how technology trends will play out, executives can plan ahead better by tracking the development of new technologies, anticipating how companies might use them, and understanding the factors that affect innovation and adoption.

To that end, we have worked with the internal and external experts on the McKinsey Technology Council to identify and interpret 14 of the most significant technology trends unfolding today (see sidebar "About the McKinsey Technology Council"). This study builds on the trend research we shared last year, adding fresh data and deeper analysis to provide a more granular assessment of trends in two thematic groups: Silicon Age, which encompasses digital and IT technologies, and Engineering Tomorrow, which encompasses physical technologies in domains such as energy and mobility.

Our analysis examines such tangible, quantitative factors as investment, research activity, and news coverage to gauge the momentum of each trend. We also conducted dozens of interviews and performed hundreds of hours of research to learn which industries are apt to benefit most as they absorb these technologies. And, recognizing that trends can speed up, slow down, or change course, we examined the uncertainties and questions surrounding each trend.

This research overview, an accompanying online interactive, and a set of 14 in-depth trend profiles lay out these considerations in more detail so that executives can better comprehend how the individual trends are developing and interacting, and what these developments might mean for their organizations. Looking across the trends, we also arrived at some general observations that leaders might find instructive.

New dynamics

First, several trends that are based on proven and mature technologies—namely applied AI, advanced connectivity, future of bioengineering, and cloud and edge computing—score more highly on quantitative measures of innovation, interest, and investment than trends based on technologies that are still in the early stages of development. These trends also tend to have viable applications in more industries, which places them closer to a state of mainstream adoption than other trends (Exhibit 1).

The next noteworthy group consists of trends that are closely aligned with sustainability priorities. These trends—which we call future of clean energy, future of sustainable consumption, and future of mobility—display increasing levels of innovation, interest, and investment. Indeed, of all the 14 trends we studied, the clean-energy and mobility trends attracted the most investment. They have also emerged as significant across multiple industries.

Outside the first two major categories, newer and less-proven digital tools power a set of emerging trends: industrializing machine learning, immersive-reality technologies, trust

architectures and digital identity, next-generation software development, and quantum technologies. These trends could have major benefits for businesses. For example, industrializing machine learning (ML) can shorten the production time frame for ML applications by 90 percent. But much work must be done to develop, refine, and commercialize the technologies that underpin these trends. As a result, it's unclear how long it will take for these trends to be adopted in multiple sectors, let alone to realize the potential seen by proponents.

About the McKinsey Technology Council

Technology is changing everything in our work and home lives. We launched the McKinsey Technology Council to help understand what is coming and how it will affect us all—taking a "look around the corner" toward the futures that technology change can unlock as well as the tough questions it raises.

We will look at a spectrum of technologies, from computing to biology, and their applications across all sectors, from mining to entertainment. We look at the science, how it translates into engineering, and when it will accelerate to impact—at scale, and around the world. We are doing this with a diverse group of more than 50 senior leaders in McKinsey, who come from a wide range of science and technology backgrounds, and orchestrating exchanges between them and dozens of leading scientists, entrepreneurs, founders, and researchers who are leading pathbreaking work in their fields.

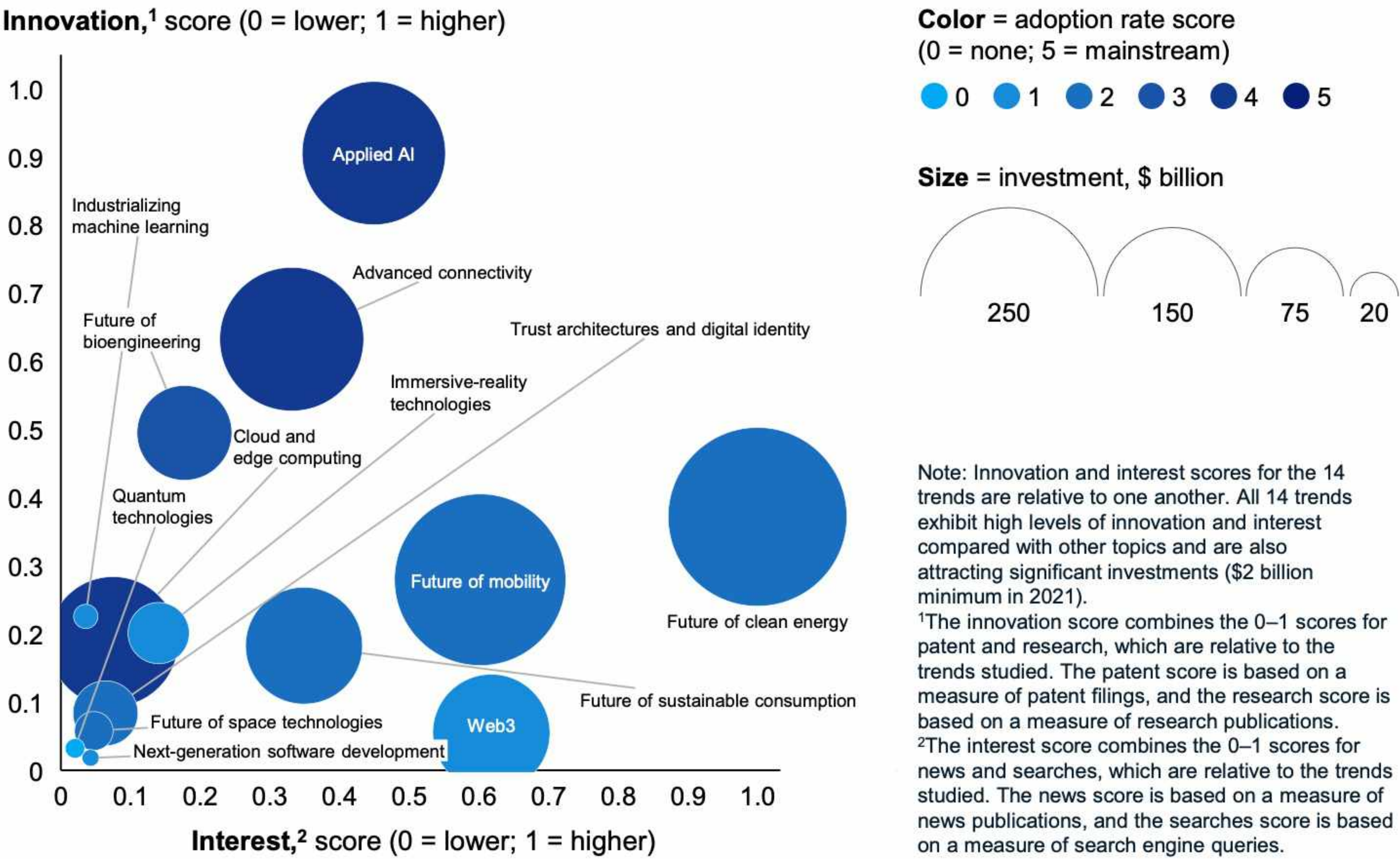
—Lareina Yee, senior partner, McKinsey; chair, McKinsey Technology Council

Introduction (continued)

Exhibit 1

Applied AI recorded the highest innovation score of all 14 trends, while clean energy drew the most interest and investment.

Innovation, interest, investment, and adoption, by technology trend, 2021



Introduction (continued)

Despite the uncertainty of these newer trends, it's apparent that industries are broadly exposed to changes resulting from technological innovation and the diffusion of new technology-enabled business practices. When we looked at how frequently news reports mentioned companies in 20 sectors alongside different trends, we found that most of these sectors display a meaningful association with five or more trends. The mature and sustainability-focused trends described above exhibit a close connection with multiple industries. Several nascent trends also show a moderately close connection with many industries. Overall, we see that press coverage connects the automotive, manufacturing, telecommunications, and technology sectors closely to at least one trend and moderately to several others (Exhibit 2). We plan to explore the effects that technology trends are having, and could have, on sectors in more depth over the coming months. (For more about how we performed this work, please see the sidebar "Research methodology.")

Looking ahead

We expect changes like these will accelerate and intensify in the years to come, much as they have during the roughly three decades since the start of the internet revolution. The changes will not only alter the competitive landscape but also exert ever-more powerful effects on society: reshaping markets, boosting productivity, spurring economic growth, and enhancing lives and livelihoods.

When we look at these trends, what impresses us more than anything else is the massive effect that technology will have on every sector. The next few decades promise to be a time in which technologies progress ever-more quickly from science to engineering to impact—at scale, and around the world. We also expect to see the multiplying effect of "combinatorial innovation," as different technologies come together in creative ways. For example, this is happening now as organizations combine different technologies to create the metaverse and the many layers that make it up.

Informed by our colleagues and the members of the McKinsey Technology Council, we intend to publish more about the effects of technology trends on particular industries, about the efforts of companies to tap into technology trends, and about the ways that business leaders can manage the implications of technology trends for their strategies, operations, and talent. We invite you to join us in understanding how technology trends evolve and influence the world, and we invite you to share your feedback with us at: techforexecs@mckinsey.com.

About the authors

Michael Chui is a McKinsey Global Institute partner in McKinsey's Bay Area office, where **Roger Roberts** is a partner and **Lareina Yee** is a senior partner.

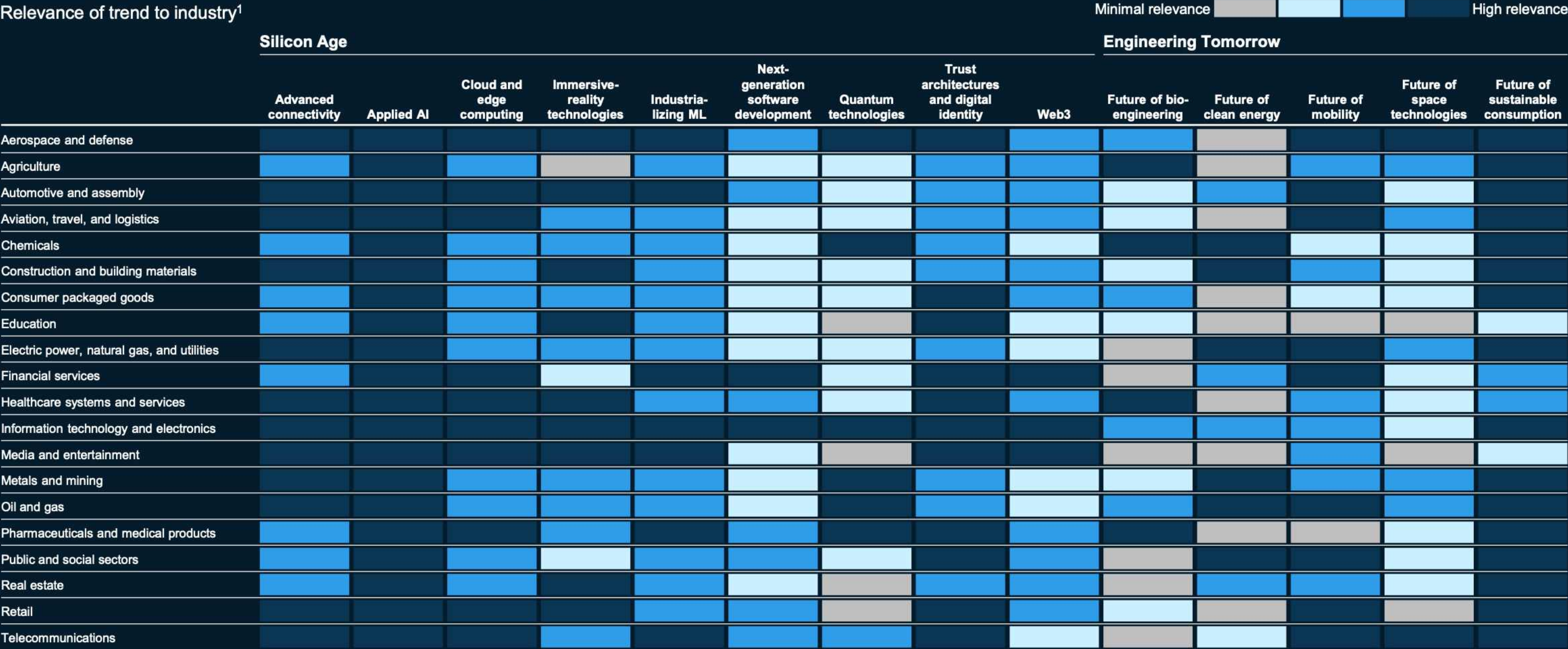
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They also wish to thank the external members of the McKinsey Technology Council.

Introduction (continued)

Exhibit 2

Most industry sectors display a meaningful association with five or more technology trends.



¹Relevance estimated qualitatively by industry experts based on trend's potential to affect an industry; degree of relevance is scaled at both trend and industry levels.

Introduction (continued)

Research methodology

To assess the development of each technology trend, our team collected data on five tangible measures of activity: search engine queries, news publications, patents, research publications, and investment. For each measure, we used a defined set of data sources to find occurrences of keywords associated with each of the 14 trends, screened those occurrences for valid mentions of activity, and indexed the resulting numbers of mentions on a 0–1 scoring scale that is relative to the trends studied. The innovation score combines the patents and research scores; the interest score combines the news and search scores. (While we recognize that an interest score can be inflated by deliberate efforts to stimulate news and search activity, we believe that each score fairly reflects the extent of discussion and debate about a given trend.) Investment measures the flows of funding from the capital markets into companies linked with the trend. Data sources for the scores include the following:

- *Patents.* Data on patent filings are sourced from Google Patents.
- *Research.* Data on research publications are sourced from the Lens (www.lens.org).
- *News.* Data on news publications are sourced from Factiva.
- *Searches.* Data on search engine queries are sourced from Google Trends.
- *Investment.* Data on private-market and public-market capital raises are sourced from PitchBook.

The associations shown on the industry heat map were derived by reviewing the news reports on each trend for references to specific companies within each of the 20 industries.

In addition, we updated the selection and definition of trends from last year's study to reflect the evolution of technology trends:

- *Trends added since last year's study:* industrializing machine learning, Web3, immersive-reality technologies, future of mobility, and future of space technologies
- *Trends not carried over from last year's study:* next-level process automation and virtualization (now considered an implication of several trends) and next-generation materials (partially represented by other trends)
- *Trends with adjusted definitions:* next-generation software development (partly covered under future of programming in 2021), trust architectures and digital identity (partly covered by trust architecture in 2021), future of sustainable consumption (disaggregated from future of clean technologies), and future of clean energy (disaggregated from future of clean technologies)
- *Trends with minor changes to definition:* applied AI, advanced connectivity (previously called future of connectivity), cloud and edge computing (previously called distributed infrastructure), quantum technologies (previously called next-generation computing), and future of bioengineering (previously called Bio Revolution)

Trends at a glance

Silicon Age

Advanced connectivity

Applied AI

Cloud and edge computing

Immersive-reality technologies

Industrializing machine learning

Next-generation software development

Quantum technologies

Trust architectures and digital identity

Web3

Engineering Tomorrow

Future of bioengineering

Future of clean energy

Future of mobility

Future of space technologies

Future of sustainable consumption

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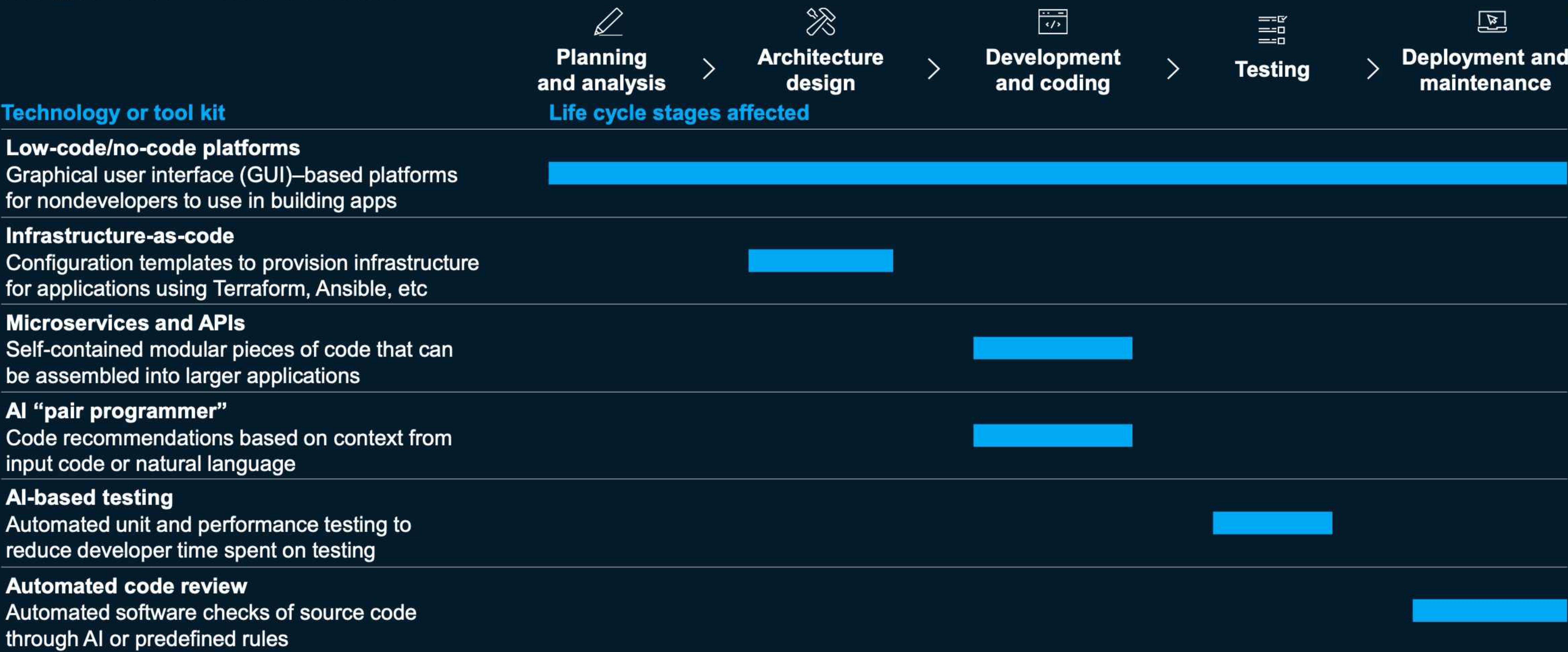
Next-generation software development

August 2022



What is this trend about?

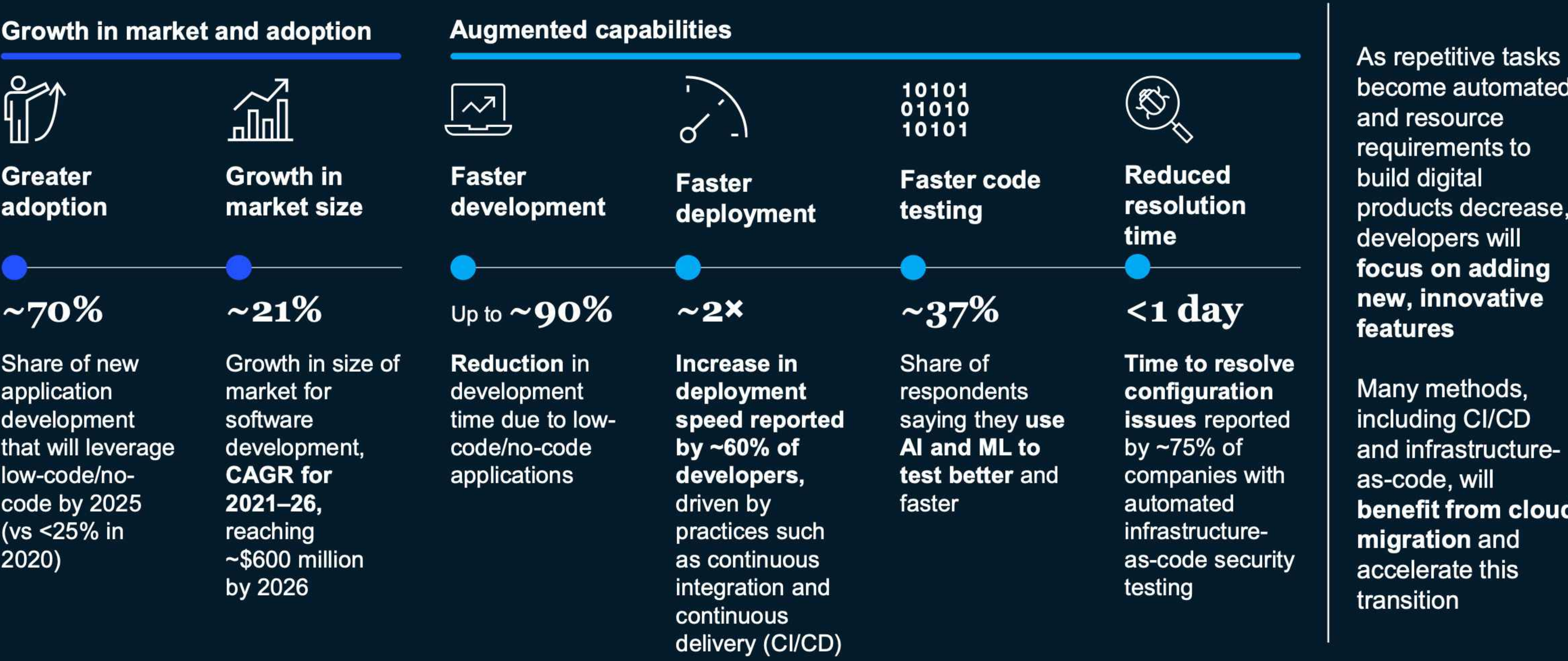
The next generation of software development involves tooling that aids in the development of software applications, improving processes and software quality across each stage of the software development life cycle, including AI-enabled development and testing, as well as low-code/no-code tools



Source: McKinsey analysis

Why should leaders pay attention?

Developers will focus more on the capabilities their applications would enable than on the details of building the apps



Source: Gartner; "Software development AI market overview," IndustryARC; *Intelligent process automation and the emergence of digital automation platforms*, Red Hat, February 2018; *A maturing DevSecOps landscape*, GitLab, May 4, 2021; Forrester Analytics Business Technographics Developer Survey, 2020; *Infrastructure as code security insights*, Snyk, 2021

Why are the technologies interesting, compared with what already exists?

From manual, time-intensive work flows and techniques ...

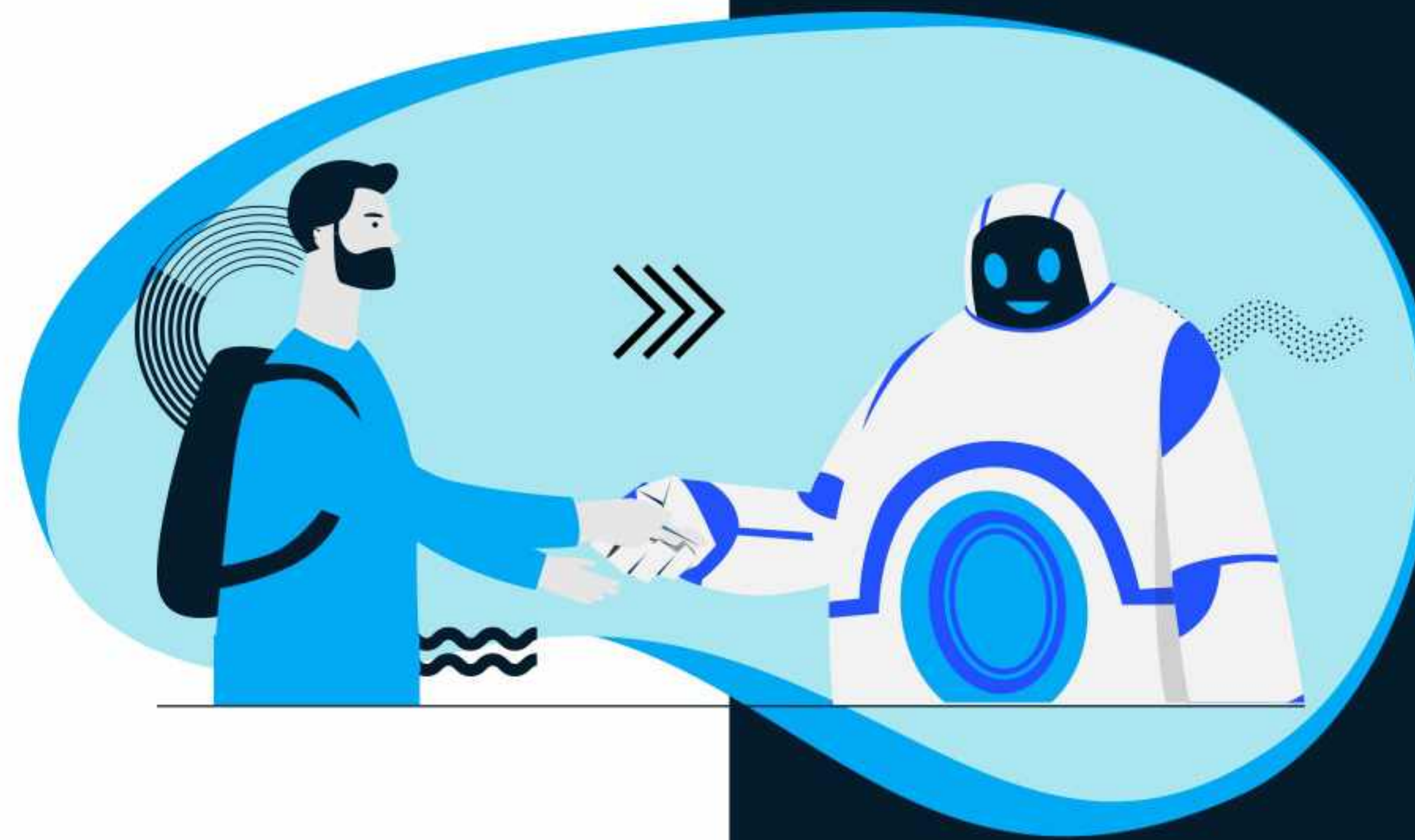
Reliance on dedicated developers to participate in every step of the development cycle, from planning to maintenance, contributes to higher costs and talent gaps

Manual infrastructure configuration and monitoring involve high mean time to restore (MTTR), security risks, and task repetition, leading to inefficient resource utilization

Developers working together to write code as 'pair programmers' on the same workstation expend a high number of person-hours to build the program

Development cycles are slow because teams experience interruptions, code has more defects, and time is spent on manual tasks

Source: McKinsey analysis



... to automated, simplified, and faster development techniques

Greater participation of 'citizen developers' (business users who have insignificant technical experience but are able to build business applications without involving technical teams) facilitates quick development of solutions more aligned to business needs

Automated configuration and monitoring through infrastructure-as-code reduces downtime and increases overall productivity and security

AI-based pair programmers are making solo developers more efficient and improving quality of code

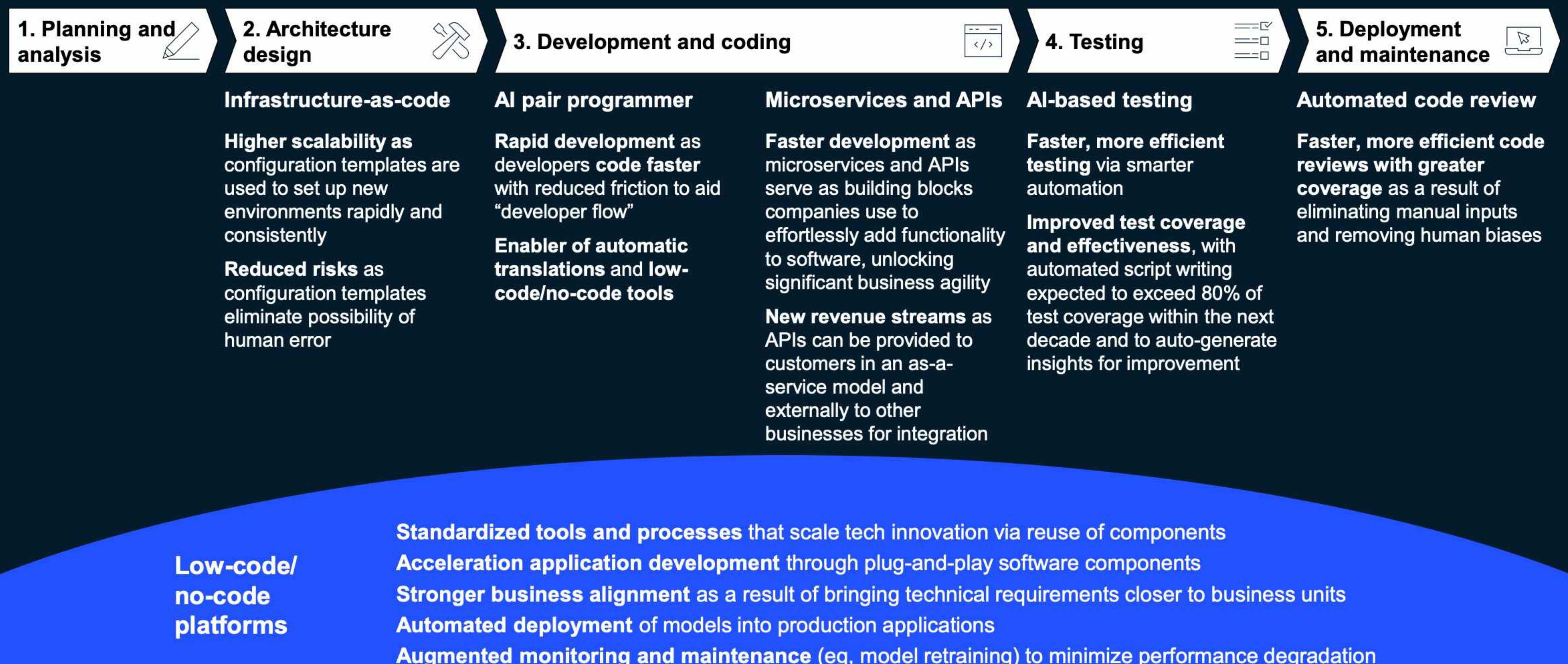
Fully automated CI/CD pipelines enable lower disruption, higher code quality, and drastically shorter development cycles

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What are the most noteworthy technologies?

Across the entire software development life cycle, technologies are already improving developer velocity

Not exhaustive



Source: Expert interviews; McKinsey analysis

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What industries could be most affected?

Beyond the information technology and electronics industry, these technologies will have an **impact on software development across all industries** by reducing digitization challenges

Many industries are already reaping the **benefits** of low-code/no-code platforms, given their **common qualities** and requirements

Source: Expert interviews; McKinsey analysis

Next-generation software development		
High industry relevanceMedium industry relevance		
Industry	Examples	Common industry qualities
 Financial services	Evolving business rules for processes such as onboarding, know your customer (KYC), and customer due diligence can be continuously handled by business analysts for efficiency	Compliance requires a wide variety of frameworks, protocols, and regulations, which typically vary by region, license agreement, etc Heavily process-based industries Significant customization requirements Rapid pace of innovation to meet evolving customer needs
 Healthcare systems and services; pharmaceuticals and medical products	Case management processes for handling customer data, tailored and specific processes for high-risk patients, development and testing of new drugs, etc, can be customized by healthcare providers	
 Manufacturing processes in automotive and assembly and aerospace and defense	Production floor management allows industrial engineers to optimize operations, reduce training expenses for new developers, reduce production floor failures, and standardize safety/handover protocols	
 Retail	Consumer-friendly front-end applications can be rapidly created and tailored to the needs of an organization and its customers	
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Who has successfully created impact with next-generation software development?

Leading players across industries have already leveraged advanced DevOps tools to optimize their SDLC¹

Stage of SDLC ¹	Technology	Example
 Architecture design	Infrastructure-as-code	Decathlon used infrastructure-as-code to automate infrastructure deployment , reducing deployment time from weeks to 30 minutes, allowing IT teams to focus on more complex tasks
 Development and coding	Automated CI/CD	Capital One leverages microservices and automated CI/CD to increase delivery speed without compromising quality through reusable building blocks and generation of templated pipelines
 Testing	AI-based test automation	Goldman Sachs uses the AI-based tool Diffblue Cover to generate unit tests for legacy software, leading to a 180× increase in the speed of writing tests for a core back-end application
 Deployment and maintenance	AI-based code reviews	Atlassian uses AI-based tools by Amazon Web Services to improve code performance by identifying code paths that demonstrate poor CPU ² utilization or latency

¹Software development life cycle.
²Central processing unit.
Source: Company websites; expert interviews

What uncertainties must be resolved for the trend to achieve scale?



Low-code/no-code platforms

Modest amount of customization is possible, compared with traditional programming languages

Monitoring and debugging applications is **difficult**, especially when they are integrated across several low-code/no-code platforms



Infrastructure-as-code

Comprehensive monitoring and version control is required to ensure errors do not spread across servers

Fragmented vendors could disrupt integrated applications, given uncoordinated changes and upgrades



AI "pair programmer"

Generated code **may be unusable or inefficient** and **may have security vulnerabilities**

Coders can be steered in the wrong direction if tools are not regularly updated with the standards or trained on clean, fast code



Microservices and APIs

Customizing APIs is difficult without significant time and effort

APIs introduce security risks by adding another attack layer that can be exploited



AI-based testing

Autonomous tools are typically specialized (eg, by programming language, test type)

Companies over-rely on automated testing/reviews when this tech scales; humans do not consistently check for errors in test and review outcomes



Automated code review

Tools do not **identify all defects and inefficiencies** in code

Source: Expert interviews; McKinsey analysis

What are some topics of debate related to the trend?

- 1

To what extent can no-code tech reduce the need for traditional software developers?

While low-code/no-code platforms help teams rapidly prototype or enable citizen developers to take over some of the work developers do, they are still not flexible enough to reduce development work at every stage of the software development life cycle (eg, when legacy systems require upgrades)
- 2

From a cultural standpoint, will teams—both developers and non-developers—embrace or resist next-generation technologies?

Automation technologies reduce time spent on development, which raises concerns for employees whose workflows are highly automatable; developers, testers, and analysts may be reluctant or eager to switch to new technologies, depending on job security, technical comfort, etc
- 3

What intellectual-property issues might affect code written by an AI application?

As companies leverage AI generation tools, there is a concern around ownership: Will the company that developed the application own it, or will it belong to the AI-enabled code generation tool provider?
- 4

To what extent will business units take responsibility for the 'health' of applications?

As next-generation software brings development capabilities to "citizen developers" embedded in business units, questions about organizational structure and responsibilities emerge—eg, as business users create applications, who is responsible for maintaining them?

Source: Expert interviews; McKinsey analysis

Additional resources

Related reading

[Developer Velocity: How software excellence fuels business performance](#)

[Security as code: The best \(and maybe only\) path to securing cloud applications and systems](#)

[Developer Velocity at work: Key lessons from industry digital leaders](#)